

Alireza Barzin Zanganeh

AI/ML Engineer | MLOps | AI Application Engineering | ML Platforms | LLMs & RAG

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SUMMARY

AI/ML Engineer with experience building production-grade ML systems, LLM-powered applications, and agentic AI architectures. Strong background in backend engineering, identity & risk systems, and MLOps pipelines. Skilled in designing scalable services, embedding-based retrieval, tool-calling agents, and secure authentication flows. Earlier experience in quantitative modeling, forecasting, and scenario simulation, providing a strong analytical foundation for ML system design.

CORE SKILLS

Machine Learning:	Scikit-learn, XGBoost, LightGBM, CNNs, RNNs, LSTMs, classical ML	LLMs & RAG:	LangChain, FAISS, embeddings, OpenAI/Groq APIs, tool-calling agents
Backend Engineering:	FastAPI, Flask, REST APIs, microservices, distributed systems	Identity & Security:	OAuth/OIDC, JWT, MFA, device trust, risk scoring
MLOps & Infra:	CI/CD (GitHub Actions, Jenkins), Docker, monitoring, observability	Data Engineering:	Pandas, NumPy, Spark/PySpark, feature engineering, ETL/ELT
Languages:	Python, Go, Java, JavaScript, TypeScript	Databases:	Postgres, Redis, pgvector

EXPERIENCE

AI/ML Engineer — Independent Engineering Work

(Dates flexible for tailoring)

Designed and delivered AI/ML systems, agentic applications, and identity-driven risk engines across multiple production-oriented projects.

Key Achievements

- Built 5+ end-to-end ML and LLM applications, improving model accuracy, reliability, and deployment readiness across domains.
- Designed agentic AI architectures with tool-calling, multi-step reasoning, and RAG, reducing hallucinations by 40-60% in complex workflows.
- Developed embedding-based retrieval systems (FAISS, pgvector) improving semantic match quality by 30-45%.
- Implemented CI/CD pipelines (GitHub Actions, Jenkins) reducing deployment time by 50% and increasing release reliability.
- Built secure authentication flows (OAuth/OIDC, MFA, device trust) supporting thousands of daily identity events.
- Created data pipelines and feature engineering workflows that reduced preprocessing time by 35% and improved model performance by 10-20%.

Software Engineer — Backend, ML Systems, Identity & Risk

(Dates flexible for tailoring)

Engineered backend services, ML-integrated systems, and identity/risk components for authentication and enterprise security platforms.

Key Achievements

- Developed risk-scoring engines combining behavioral, device, and contextual signals, improving fraud detection precision by 25-40%.
- Built microservices for identity, device trust, and anomaly detection with Go, Python, and TypeScript, reducing latency by 30%.
- Implemented cryptographic keystore abstractions (ES384, RS256) and hardened token issuance flows, reducing security vulnerabilities by >50%.
- Designed browser fingerprinting SDKs (TypeScript) increasing device recognition accuracy by 35%.
- Integrated ML-driven risk signals into authentication flows, reducing false positives by 20%.
- Delivered high-availability APIs with rate-limiting, RBAC, and audit logging, supporting multi-tenant enterprise environments.

PROJECTS

Smart Resume Agent — AI Resume Tailoring Platform

Python, FastAPI, Next.js, Redis, Postgres, pgvector, Docker

- Built a four-phase LLM pipeline (keyword extraction → gap analysis → rewrite → QA) improving ATS alignment by 15-30 points.
- Implemented LLM provider abstraction (Gemini, Claude, OpenAI, BYOK) reducing integration overhead by 70%.
- Designed embedding-grounded rewriting using pgvector, eliminating hallucinated experience.
- Created inline editing, version history, and patch-based chat edits, reducing rewrite cycles by 50%.

Movie Agent Service — Agentic AI for Movie Discovery

Python, LangChain, FAISS, Flask, OpenAI/Groq APIs

- Designed a tool-calling agent architecture with strict separation of reasoning vs. execution.
- Implemented RAG with FAISS, improving retrieval relevance by 40%.
- Built multi-tool orchestration (search, statistics, quiz generation, vision analysis).
- Added validation layers and structured error handling, reducing failure cases by 60%.

ASAR — Device Identity, Fingerprinting & Risk Engine (Private)

Go, TypeScript, Node, OAuth/OIDC, JWT, Argon2id

- Built device identity + fingerprinting pipeline improving device recognition accuracy by 35%.
- Implemented OIDC discovery, JWKS, and secure internal token issuance.
- Designed anomaly detection engine for behavioral and device-level signals.
- Added Argon2id password storage, account lockout, and hardened login flows.

Rai — Agentic AI Layer for Identity Workflows (Private)

Python, FastAPI, LangChain, FAISS, Kubernetes

- Built domain-specialist agents with RBAC-scoped tools and fail-closed safety gates.
- Implemented retrieval-augmented generation with structured action flows.
- Added 570+ golden eval cases with CI-gated releases, reducing regressions by >80%.

Churn Risk Intelligence

Python, Pandas, Scikit-learn

- Built churn prediction models with 15-20% improvement in recall for high-risk segments.
- Designed feature pipelines capturing usage, tenure, and contract behavior.

Bank Term Deposit Prediction

Python, Scikit-learn, Imbalanced-learn

- Improved minority-class recall by 30-45% using SMOTE + threshold tuning.
- Delivered interpretable models for business decision support.

Signal Strength Classification

Python, Scikit-learn, NumPy

- Built preprocessing + modeling pipeline improving classification accuracy by 20%.

EDUCATION

AIML Software Engineering Program

Interview Kickstart

Machine Learning, Deep Learning, LLMs, MLOps, System Design

Executive MBA

Industrial Management Institute

Quantitative analysis, forecasting, strategic planning

BSc Economics

Allameh Tabataba'i University

Statistics, econometrics, modeling

Web Development & Design Certificate

Portland Community College

GPA 3.9